

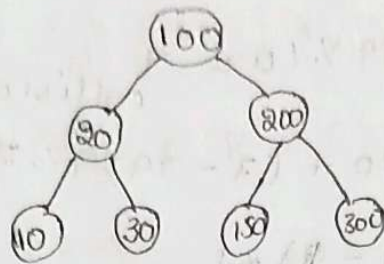
CO3-AT3 - Tree Traversal challenges

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Redd

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1)

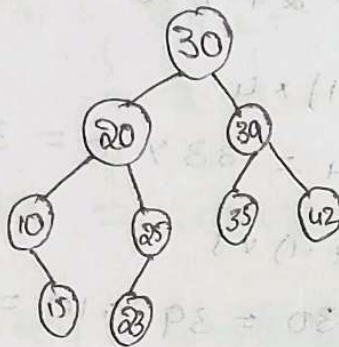


Preorder: 100, 20, 10, 30, 200, 150, 300 (NLR)

Inorder: 10, 20, 30, 100, 150, 200, 300 (LNR)

Post order: 10, 30, 20, 150, 300, 200, 100 (LRN)

2)

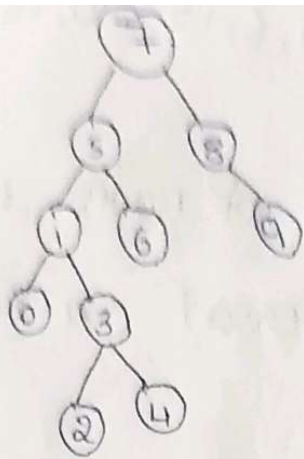


Inorder(LNR): 10, 15, 20, 23, 25, 30, 35, 39, 42

Pre order(NLR): 30, 20, 10, 15, 25, 23, 39, 35, 42

Post order(LRN): 15, 10, 23, 25, 20, 35, 42, 39, 30

3) Assume the digits 7, 5, 1, 8, 3, 6, 0, 9, 4, 2 are added in sequence into a binary search tree that is initially empty. what is the resulting tree's traversal sequence

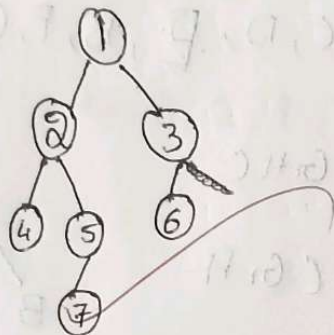


Preorder (NLR): 7, 5, 1, 0, 3, 2, 4, 6, 8, 9

Inorder (LNR): 0, 1, 2, 3, 4, 5, 6, 7, 8, 9

Postorder (LRN): 0, 2, 4, 3, 1, 6, 5, 9, 8, 7

5)

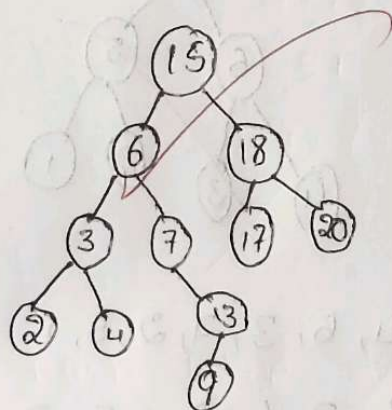


Inorder (LNR): 4, 2, 7, 5, 1, 3, 6

Preorder (NLR): 1, 2, 4, 5, 7, 3, 6

Postorder (LRN): 4, 7, 5, 2, 6, 3, 1

6)

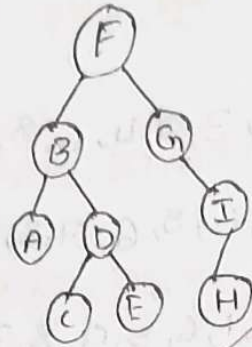


Inorder: 2, 3, 4, 6, 7, 9, 13, 15, 17, 18, 20 (LNR)

Preorder: 15, 6, 3, 2, 4, 7, 13, 9, 18, 17, 20 (NLR)

Post order: 2, 4, 3, 9, 13, 7, 6, 12, 20, 18, 15 (LRN)

7) what is the sequence of nodes when applying in-order traversal on the binary tree

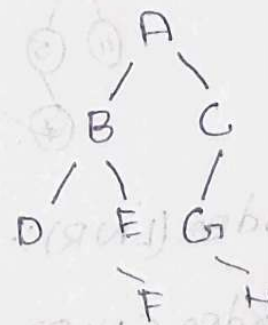


Inorder (LNR): A, C, D, E, B, F, G, H

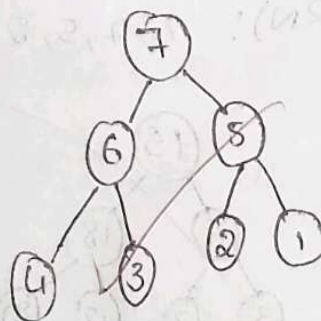
8) Inorder - D B E F A G H C

Preorder - A B D E F C G H

Postorder - D F E B H G C A
(LRN)



9)

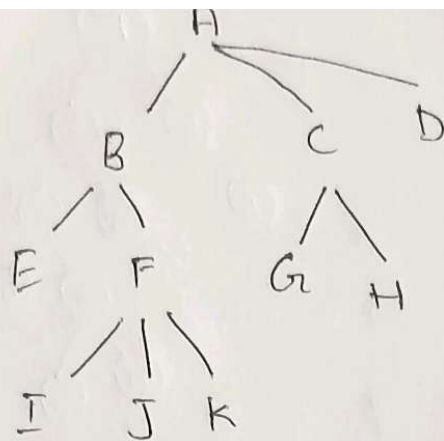


Inorder (LNR) : 4, 6, 3, 7, 2, 5, 1

Postorder (LRN) : 4, 3, 6, 2, 1, 5, 7

Preorder (NLR) : 7, 6, 4, 3, 5, 2, 1

10)



pre order: A, B, E, F, I, J, K, C, G, H, D

post order: E, I, J, K, F, B, G, H, C, D, A

level order: A, B, C, D, E, F, G, H, I, J, K.

Ans